

Durga Charan Potukuru

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PROFESSIONAL SUMMARY

Senior Software Engineer with deep experience designing and delivering enterprise-scale web applications. Currently lead the architecture and engineering of StrasZ Assessment Systems' CPA exam delivery platform. Lead engineering teams, apply Domain-Driven Design and micro-frontend architecture across Angular and .NET Core ecosystems, and align service boundaries with business capabilities in an Agile setting.

CORE COMPETENCIES

Architecture & Design: Composite UI / Micro-frontends, Autonomous Components, Microservices, SOA, Domain-Driven Design (DDD), CQRS, Event Sourcing, Sagas / Long Running Processes, Eventual Consistency, Read Models, Distributed Systems, SOLID, Clean Architecture

Frontend: AngularJS, Angular 2+ / Angular 18, TypeScript, RxJS, MonoRepo (Nx-style), React, Redux, Vue, Webpack, NPM, HTML5, CSS3, SASS / LESS, Bootstrap

Backend & Messaging: .NET Framework / .NET Core, C#, Node.js, Express.js, RESTful APIs, Azure Service Bus, RabbitMQ, NServiceBus patterns

Cloud & Serverless: Azure App Service, Azure Functions (serverless), Azure AD, Azure Key Vault, Azure Cognitive Search, Synapse Analytics, Azure Data Lake Storage, Cosmos DB

DevOps & CI/CD: Azure DevOps, GitHub Actions, Jenkins, Docker, Kubernetes, Git, BitBucket, NPM, Yarn, NuGet, Webpack pipelines

Quality & Methodology: Test-Driven Development (TDD), Behavior-Driven Development (BDD), Pair Programming, Code Review, Event Storming, Agile / Scrum, Continuous Integration / Continuous Delivery

Databases: Cosmos DB, MongoDB, RavenDB, Microsoft SQL Server

Domain: Assessment & Computer-Based Testing (CPA exam delivery), Healthcare, Enterprise SaaS

PROFESSIONAL EXPERIENCE

Senior Software Engineer

US Software Group Inc. | 06/2020 – Present

Client engagement: StrasZ Assessment Systems, Exam Management Suite (CPA exam platform)

- Lead an 8-engineer team building StrasZ's Exam Management Suite. Jointly own the enterprise-wide architecture runway and maintain design integrity through code reviews, design workshops, and architectural decision records.
- Identify skill gaps on the team, build learning plans, and run event storming workshops to apply DDD to the microservices roadmap. Mentor engineers on TypeScript, Angular architecture, and .NET.
- Architect and implement the front-end on Angular 18 and TypeScript across a monorepo, defining shared libraries, module boundaries, lazy-loading, and reusable components used across multiple products.
- Designed a Composite UI / micro-frontend architecture so feature modules like the Excel experience, search, and exam navigation can be built and deployed independently and composed into the CPA exam shell.
- Apply Domain-Driven Design to align service boundaries with business capabilities such as item authoring, exam delivery, and scoring, which has reduced cross-team coupling and clarified ownership.
- Designed distributed, high-throughput exam delivery flows using CQRS with separate read models and Sagas for stateful, long-running exam sessions, with eventual consistency between authoring and delivery contexts. The platform supports hundreds of thousands of exam deliveries each year.

- Use Azure Service Bus for messaging between microservices, including topics, subscriptions, dead-lettering, and retry policies, applying patterns similar to NServiceBus and RabbitMQ.
- Build backend microservices in C# on .NET Core with bounded-context APIs and Cosmos DB persistence, holding the .NET codebase to SOLID principles and clean code standards.
- Modernize parts of the platform to Azure Functions for event-driven processing, scheduled jobs, and lightweight integrations, reducing infra overhead.
- Set up CI/CD pipelines on Azure DevOps and GitHub Actions for the monorepo using Git, NPM, Webpack, Docker, and Kubernetes. Pipelines run linting, unit and integration tests, and gated releases.
- Use TDD and BDD extensively across the team, with pair programming, test-first workflows, and a shared definition of done. Code quality and regression rates have both improved noticeably.
- Improved CPA exam runtime performance by 7% through front-end rendering optimization, bundle and lazy-load tuning, change-detection refinement, and back-end query and caching work.
- Built the embedded Excel spreadsheet experience inside the CPA exam using GrapeCity SpreadJS, including formula evaluation, item-state persistence, and accessibility. This is critical for financial simulation items.
- Designed a custom search capability on Azure Cognitive Search with a custom analyzer tuned to the assessment domain, enabling sub-second item lookup over large content stores.
- Work in an Agile/Scrum setting, delivering value sprint by sprint, contributing to internal design reviews, and pushing for continuous improvement of architecture, tooling, and delivery practices.

Software Engineer

Marlabs Inc. | 01/2019 – 06/2020

Client engagement: AmeriHealth Caritas

- Led the client-side team building HIPAA-compliant Single Page Applications using React, Redux, React Router, Bootstrap, and HTML5/CSS3. Built a reusable component library that standardized the UI across multiple internal portals.
- Established front-end patterns including container/presentational separation, normalized Redux state, async middleware with redux-thunk, and memoized selectors to keep large data grids performant under heavy load.
- Built RESTful APIs and Node.js / Express.js services for member, provider, and document workflows. Improved backend response times by about 30% through query optimization, caching, payload trimming, and connection pooling.
- Designed a custom electronic-signature workflow on top of Adobe Sign, covering template management, signer routing, status webhooks, and audit trails to meet compliance requirements.
- Implemented authentication and role-based authorization across services using JWT and OAuth2, and integrated with enterprise identity providers for single sign-on.
- Wrote unit and integration tests with Jest, Enzyme, and Mocha. Took part in code reviews and CI pipeline configuration, which helped raise code coverage and reduce production issues.
- Worked closely with product owners, UX, and QA throughout the SDLC to refine requirements, validate prototypes, and meet enterprise quality and accessibility standards.
- Supported production releases and incident triage. Performed root-cause analysis across the React front-end, Node services, and database layer, and drove permanent fixes back into the codebase.

Full Stack Developer

Ramky Enviro Engineers Ltd | 08/2014 – 07/2017

- Delivered features end to end across two enterprise web applications, owning everything from database schema and REST API design through to the front-end UI.
- Built responsive front-ends with AngularJS, jQuery, Bootstrap, HTML5, CSS3, and JavaScript (ES5/ES6), introducing modular controllers and services that improved maintainability.
- Designed and implemented RESTful APIs on Node.js and Express.js, including request validation, error handling, structured logging, and consistent JSON contracts for both web and mobile clients.

- Implemented authentication and authorization using PassportJS and JWT, including session management, refresh-token flows, and role-based access control.
- Built data models on Microsoft SQL Server and MongoDB. Wrote optimized queries, stored procedures, and aggregation pipelines, and tuned indexes to handle growing transactional volume.
- Built custom dashboards and visualizations with D3.js and Highcharts.js, including time-series charts, geo-maps, and KPI tiles, which improved operational visibility and reporting by an estimated 40%.
- Developed an internal continuous-integration utility to streamline build, test, and deployment, which reduced manual release effort.
- Used Git with pull-request reviews and took part in Agile ceremonies including stand-ups, sprint planning, and retrospectives, contributing to estimation and refinement.
- Worked with business analysts, QA, and operations to translate domain requirements into technical solutions, and supported production deployments and post-release monitoring.

EDUCATION

Master of Science, Applied Computer Science

Northwest Missouri State University, USA

CERTIFICATIONS & RECOGNITION

- Oracle Certified Associate, Java SE 7 Programmer.
- Technical Paper Presentation, Infoquest 2014.
- Member: Excellence Leadership Academy; International Association of Neuro Linguistic Psychology.

PERSONAL PROJECTS

- Socio Analyzer (2018). Sentiment-analysis pipeline over a Twitter stream (#metoo) using Python, NLTK, and TextBlob. Achieved 79% classification accuracy.
- Cloud + Fog Computing Latency Study. Demonstrated a 9% latency improvement for cloud workloads by introducing a fog-computing tier with decoy-technology-based portability and security.
- Conference Registration System. Full-stack SPA built on Angular, MongoDB, Express, and Node.js (MEAN).
- Northwest Theatre Ticket Booking. SPA built on React, React Router, Redux, Bootstrap 4, SASS, and Webpack.